Tiktok vs Youtube's performance in 2025

I. Introduction and background

For this project, I analyzed the performance of TikTok and YouTube in 2025, two of the most influential short-form video platforms today. As both platforms have experienced rapid growth in recent years, this analysis explores what differentiates them and how their performance compares across key metrics.

Using a global dataset, I examined how video content performs across different countries, which content categories generate the highest engagement, and how growth and audience interaction vary over time. The analysis focuses on geographic, temporal, and engagement-based dimensions, including likes, shares, and comments. Through this exploration, the project aims to highlight key similarities and differences in platform performance and provide insight into how each platform enga

II. Datasets

The primary dataset used in this project is ***YouTube & TikTok Trends 2025***, sourced from Kaggle and published by **Tarek Masryo** under the Creative Commons public domain. The dataset was selected for its comprehensive coverage of platform performance in 2025, including engagement metrics, content categories, and video volume.

The dataset contains **58 variables** and covers **over 100 countries**, enabling analysis across multiple dimensions. In addition to the main dataset, several sub-files were used, including geographic summaries (`country_platform_summary`) and creator-level metrics

(`top_creators_impact`). Variables include both numerical metrics—such as engagement rate per 1,000 views, like counts, and share counts—and categorical attributes such as content category, country, and platform.

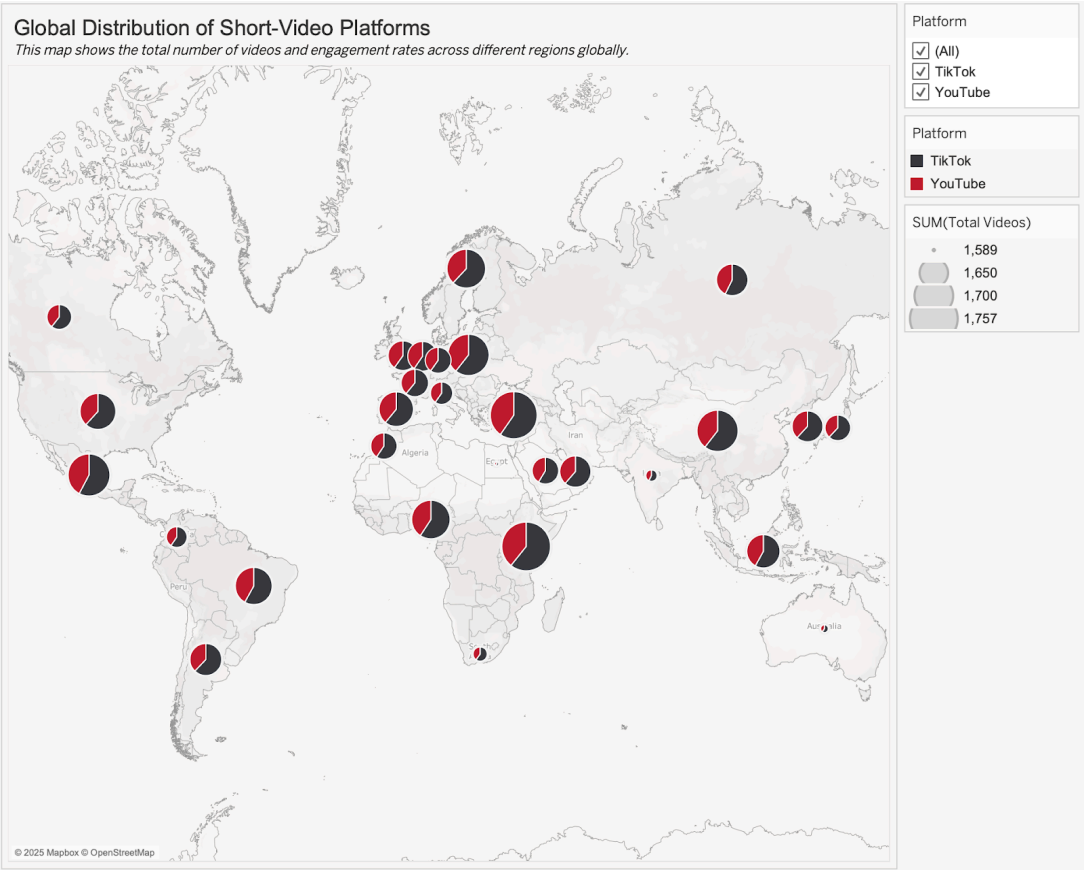
While the dataset is derived from publicly available sources rather than proprietary platform data, which may introduce limitations such as missing or imprecise values, its breadth and structure provide sufficient depth to support meaningful comparative analysis across platforms and regions.

III. Data story

The first visualization examines **global video distribution and audience engagement** to understand how content performance varies across countries. This analysis uses the `country_platform_summary_2025` sub-dataset, which includes country, platform, total video volume, and engagement rate per 1,000 views.

A geographic map with an engagement heatmap was used to effectively display global patterns and regional differences. Average engagement rate was encoded using color intensity, while bubble size was scaled to represent total video volume, allowing for simultaneous comparison of reach and audience interaction.

This visualization reveals that **Kenya, China, and Mexico** are among the most active regions in terms of content creation, while **the United States, Kenya, and Japan** demonstrate some of the strongest audience engagement. Together, these patterns highlight meaningful regional differences in both content supply and user interaction across platforms.



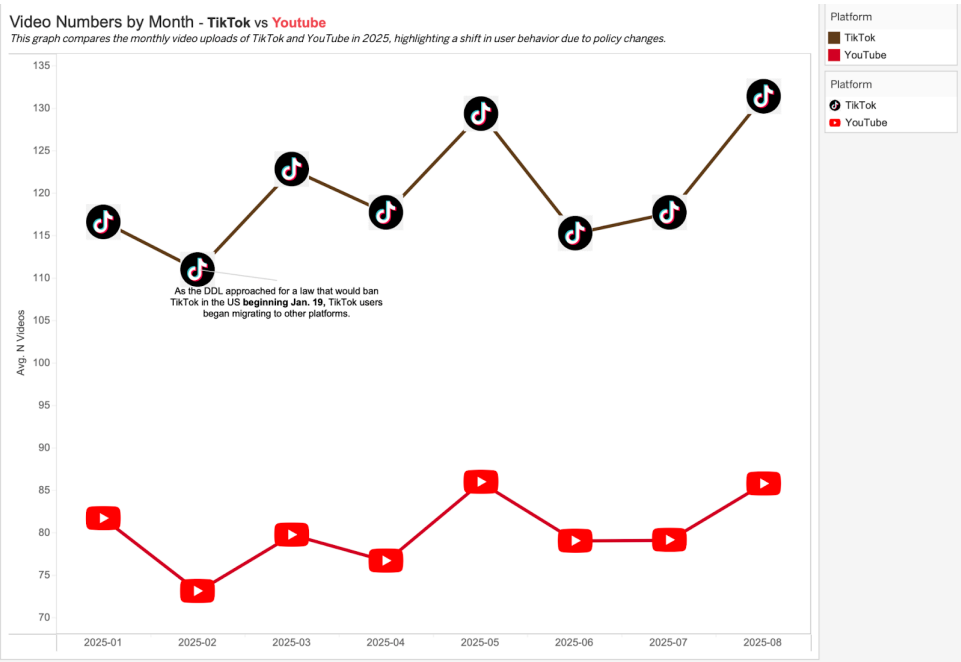
Map — Global Video Distribution & Engagement Heatmap

The next visualization uses a **line chart** to examine monthly video production trends in 2025 and compare differences between TikTok and YouTube over time. A line chart was selected to clearly highlight trends, spikes, and declines across both platforms at the same points in time, making it well suited for temporal comparison.

This visualization is based on the `Monthly_trends_2025` sub-dataset and includes **time (year–month)**, **number of videos**, and **platform** as key variables. Platform-specific colors and logos were incorporated to improve readability and enhance visual engagement.

One limitation of this dataset is that it only covers the period from **January to August 2025**, rather than a full calendar year. Despite this constraint, the analysis reveals that both platforms generally experienced **similar timing in spikes and declines**, suggesting shared external or seasonal influences. However, **TikTok exhibited higher video volume and faster growth periods** throughout the observed timeframe.

Additionally, this chart provides insight into potential **seasonal or policy-driven shifts** in content production, such as noticeable changes in TikTok activity following reports of a potential platform ban announced on **January 19, 2025**.

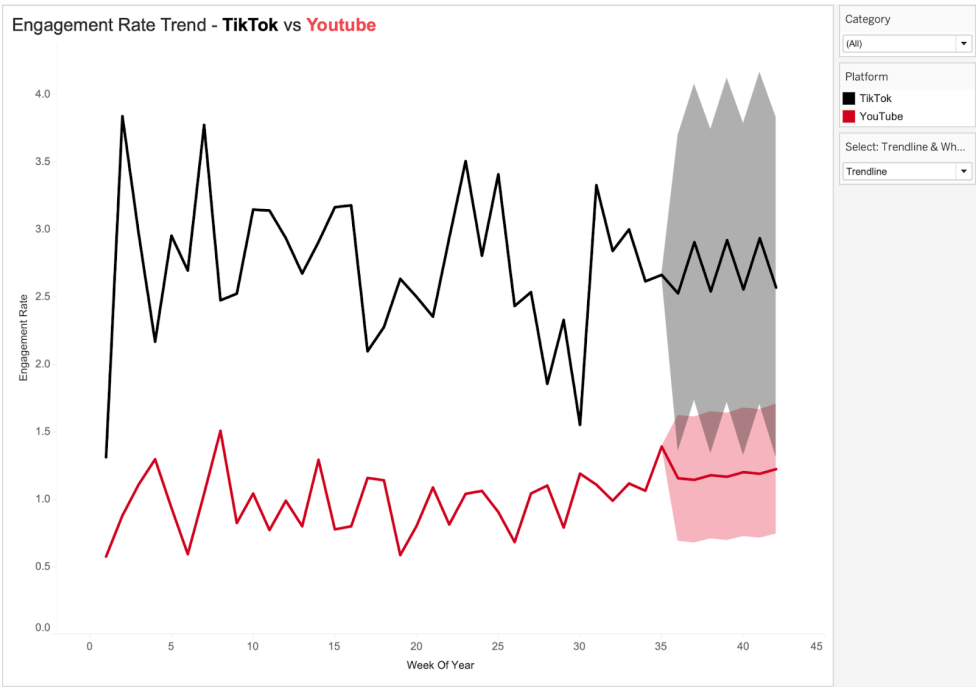


Line chart — Video numbers by Month

The next visualization uses a **line chart** to track weekly engagement trends across different video content categories and to explore potential future changes in engagement. This chart was designed to identify **seasonal or periodic patterns** in audience interaction over time.

Using the `Youtube_shorts_tiktok_trends_2025` sub-dataset, the analysis incorporates **week of year**, **engagement rate** (derived from likes, shares, and comments), **content category filters**, and **platform**. A line chart was selected for its ability to clearly illustrate engagement trends over time, while **shaded regions** were added to represent predicted engagement ranges and highlight potential future trajectories.

The analysis shows that **TikTok consistently maintains higher engagement rates than YouTube** across content categories, reinforcing its stronger audience interaction throughout the observed period.



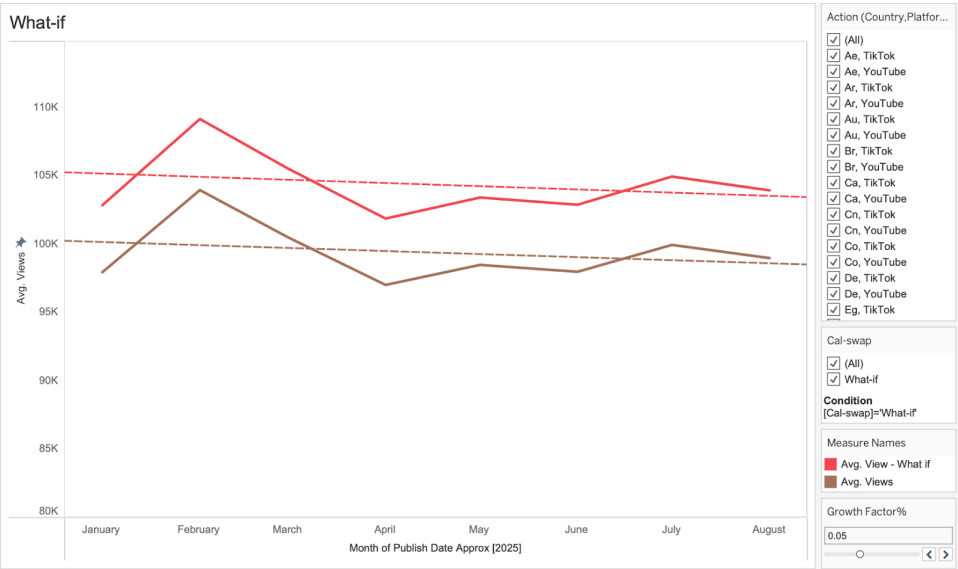
Line chart (trend analysis) — Engagement rate trend

To further explore platform performance, a **what-if analysis** was conducted to evaluate how changes in audience growth rates could impact total viewership. Using the

`Youtube_shorts_tiktok_trends_2025` sub-dataset, this visualization incorporates **time**, **average views**, **platform**, and a **hypothetical growth factor ranging from −50% to +100%** to simulate alternative growth scenarios.

Actual view trends are represented using **solid lines**, while **dotted lines** illustrate projected outcomes under different growth assumptions, allowing for clear comparison between observed and simulated performance. An interactive **growth-rate parameter** was included to enable dynamic exploration of how varying growth rates affect overall viewership.

The results demonstrate that even modest changes in audience growth can lead to **significant differences in total views for** both platforms. While TikTok shows higher volatility and sensitivity to growth changes, **YouTube exhibits more stable and consist**ent growth patterns, suggesting greater resilience under fluctuating audience conditions.

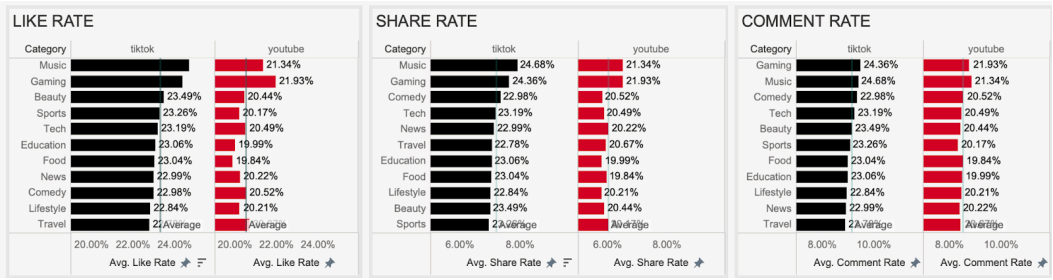


What-if analysis — Simulated view growth

The final set of visualizations uses **grouped horizontal bar charts** to compare **like, share, and comment rates** across TikTok and YouTube, segmented by content category. This analysis draws from the `Youtube_shorts_tiktok_trends_2025` sub-dataset and incorporates **platform**, **content category**, and **average engagement metrics** to evaluate differences in audience interaction.

Grouped horizontal bar charts were selected to clearly highlight contrasts between platforms and make category-level comparisons easy to interpret at a glance. The results show that **TikTok consistently outperforms YouTube across all three engagement metrics**, indicating stronger audience interaction across categories.

Across both platforms, **music and gaming** emerge as the most popular content categories, generating the highest levels of engagement. These insights offer clear value for product and content strategy, helping platform stakeholders identify strengths, uncover gaps, and prioritize categories that can most effectively drive user engagement and growth.

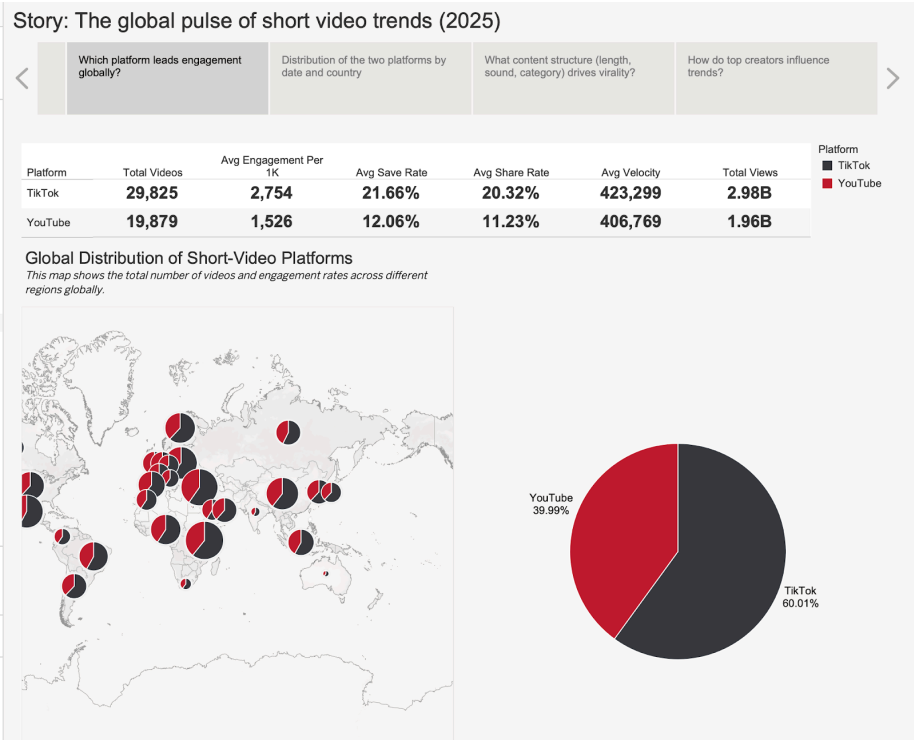


Bar charts — Category-level engagement comparison

To unify the analysis and address the research questions holistically, a **storyboard** was created to connect all visualizations into a cohesive narrative. This approach allows individual insights to be viewed within a broader context, making it easier to compare how TikTok and YouTube perform across multiple dimensions.

Additional supporting visualizations were developed, including a **bar chart** illustrating platform popularity by country using total video volume and engagement, and an **area chart** showing how top creators' view counts grow or decline over time. Together, these charts reinforce key findings and provide deeper insight into platform dynamics.

By integrating all visual elements into a single narrative flow, the storyboard highlights how the two platforms differ in growth, engagement, and creator performance, offering a comprehensive view of their relative strengths and performance patterns.



Storyboard

IV. Summary and conclusions

Across all visualizations and analyses, the central question of this project was how performance differs between TikTok and YouTube. Supported by creators across diverse content categories, styles, and trends, both platforms have established themselves as leading forces in the global short-form video ecosystem.

The analysis shows that while **YouTube demonstrates steady and consistent growth over time**, **TikTok achieves higher audience engagement and faster creator-driven growth**, fueled by a greater volume of content uploads. Geographic, temporal, and engagement-based analyses, combined with predictive what-if modeling, consistently indicate that TikTok outperforms YouTube in terms of audience interaction.

Overall, these findings suggest that TikTok currently offers creators stronger opportunities for engagement and visibility, making it a more attractive platform for driving interaction and audience growth.

References

Lee, R. A. (2025, August 9). *YouTube shorts vs. Tiktok Usage Statistics 2025: Key insights • SQ magazine*. SQ Magazine. https://sqmagazine.co.uk/youtube-shorts-vs-tiktok-usage/

Masryo, T. (2025, September 16). *YouTube & Tiktok Trends*. Kaggle. https://www.kaggle.com/datasets/tarekmasryo/youtube-shorts-and-tiktok-trends-2025